

1. What is k-NN and is it supervised or unsupervised?

k-NN is a method that predicts a value or class by looking at the closest data points. It is a **supervised learning algorithm**.

2. What is SVM and what are some limitations?

SVM finds the best line or boundary to separate different groups of data.

Limitations:

- Hard to understand
 - Needs tuning
 - Slow for large datasets
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3. How do decision trees work and how do you avoid overfitting?

Decision trees split data step-by-step based on conditions to make predictions.

To avoid overfitting:

- Limit depth
 - Prune the tree
 - Use more data
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4. Define bootstrap aggregating and ensembling in random forests

Bootstrap aggregating (bagging) means creating many datasets by sampling data randomly. Ensembling means combining many decision trees to make better predictions.

5. What is backpropagation and when do neural nets work best?

Backpropagation is a method used to update weights in a neural network by fixing errors.

Neural networks work best when:

- There is a lot of data
 - Problems are complex
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6. Define epsilon-decreasing strategy in A/B testing

It means starting by testing all options equally, then slowly choosing the best option more often over time.

7. 7 Glossary terms (simple)

1. Overfitting – model learns too much noise
2. Underfitting – model is too simple
3. Feature – input variable
4. Model – system that makes predictions
5. Training data – data used to learn
6. Test data – data used to check accuracy
7. Accuracy – how correct the model is